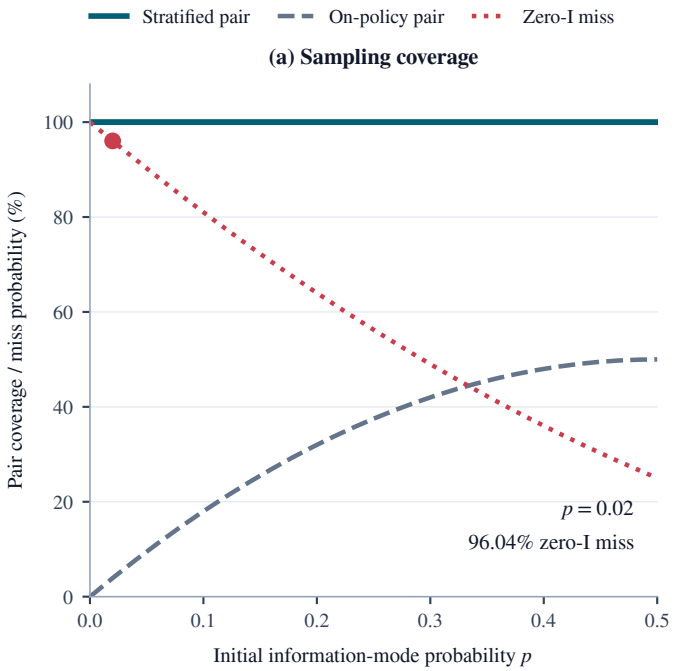
## RQ3 | Sampling coverage and gradient agreement




**(b) Gradient agreement**

Exact objective: 58/125 = 0.4640

| Estimator | $g_1$ | $g_2$ | $g_3$ |
|---|---|---|---|
| Analytical | +0.03840 | +0.04480 | +0.00000 |
| Stratified exact | +0.03840 | +0.04480 | +0.00000 |
| RLOO exact | +0.03840 | +0.04480 | +0.00000 |
| Autograd | +0.03840 | +0.04480 | +0.00000 |
| Stratified MC | +0.03814 | +0.04439 | -0.00031 |
| RLOO MC | +0.03736 | +0.04529 | -0.00164 |

Variance identity: LHS = RHS = 0.0164736

Exact two-stage setting. MC: 10,000 episodes. Gradient coordinates follow the stored parameter-vector order.